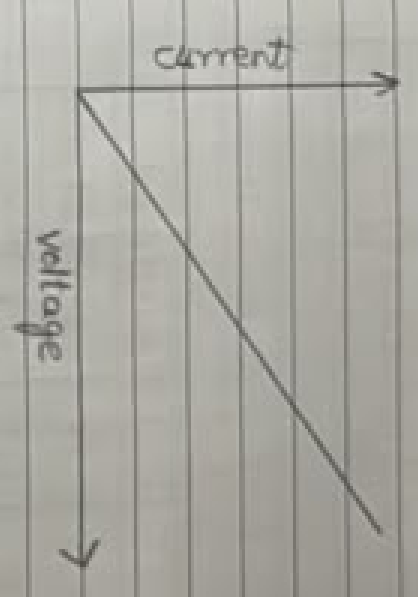


MODEL GRAPH:-

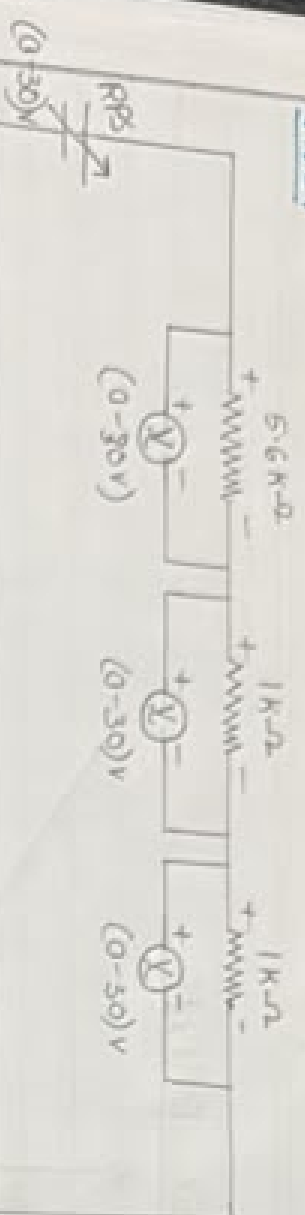


Result:-

Thus, the Ohm's Law is verified for the given circuit.

Teacher's Signature _____

KVL:



Supply Voltage	V_1	V_2	V_3	V_1	V_2	V_3
10	4.5	4.5	1.0	4.5	4.5	0.8
20	9.1	9.0	1.7	9.1	9.1	1.6
30	13.72	13.72	2.5	13.72	13.72	2.45

Calculation:

$$R_T = 5.6 + 5.6 + 5 = 12.2, \quad V_T = 10$$

$$I_T = 10/12.2 = 0.81 \text{ mA}, \quad I_T = 10/12.2 = 1.6 \text{ mA}$$

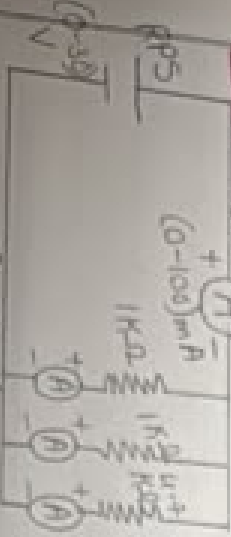
$$V_1 = 0.81 \times 10^{-3} \times 5.6 \times 10^3 = 4.536$$

$$V_2 = 0.81 \times 10^{-3} \times 5.6 \times 10^3 = 4.536$$

$$V_3 = 0.81 \times 10^{-3} \times 5 \times 10^3 = 4.05$$

$$V_1 + V_2 + V_3 = 13.72$$

KCL:



KCL calculation:

$$I = V/R$$

$$I_1 = \frac{2.7}{1} = 2.70$$

$$I_2 = \frac{2.7}{1} = 2.70$$

$$I_3 = \frac{2.7}{1} = 2.70$$

$$I_1 + I_2 + I_3 = 8.1$$

$$I_1 = 2.7, I_2 = 2.7, I_3 = 2.7$$

$$I_1 = 2.7, I_2 = 2.7, I_3 = 2.7$$

$$I_1 = 2.7, I_2 = 2.7, I_3 = 2.7$$

$$I_1 = 2.7, I_2 = 2.7, I_3 = 2.7$$

$$I_1 = 2.7, I_2 = 2.7, I_3 = 2.7$$

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$$I_1 = 2.7, I_2 = 2.7, I_3 = 2.7$$

$$I_1 = 2.7, I_2 = 2.7, I_3 = 2.7$$

$$I_1 = 2.7, I_2 = 2.7, I_3 = 2.7$$

Exp. No. 03

Date: 03

1b) Verification of Kirchhoff's Law

1. To verify Kirchhoff's Voltage Law
2. To verify Kirchhoff's Current Law

Apparatus Required

S.No.	Equipment	Type	Range	Qty
1.	RPS	-	(0-15)V	1
2.	Bread Board	-	-	1
3.	Ammeter	mc	(0-10) mA	1
4.	Voltmeter	mc	(0-10) V	1
5.	Resistor	-	1kΩ, 2kΩ, 4kΩ	each 1
6.	Connecting wires	-	-	as required

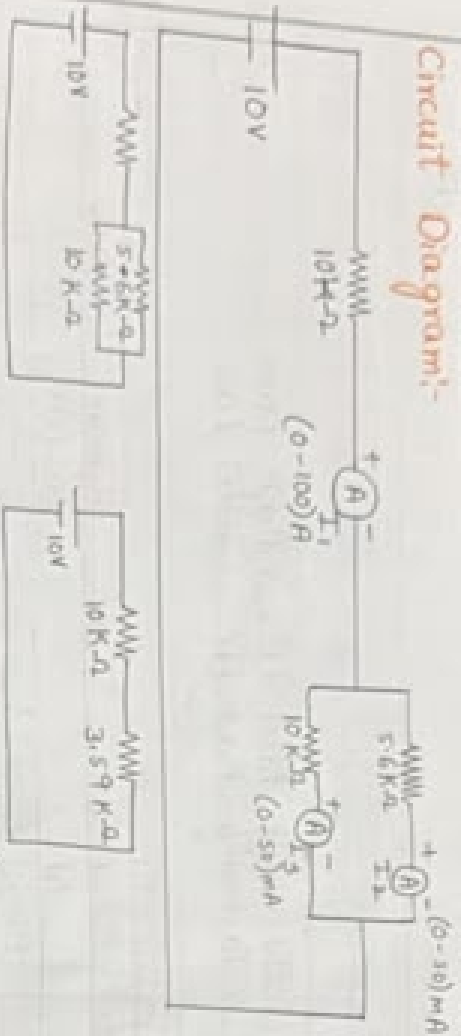
Procedure:

1. Connections are given as per the circuit Diagram.
2. Apply DC voltage to circuit from RPS.
3. Tabulate the voltmeter and ammeter readings.
4. Increase the voltage step by step to get different readings until the voltage reaches 15V.
5. Repeat step 3 for different values.
6. Switch off the power supply after bringing RPS to the minimum voltage positions.

Result: Thus, the Kirchhoff's voltage and current law is verified for the given circuit.

Teacher's Signature

Circuit Diagram:



$$I_1 = I_T = \frac{V}{R_T}$$

$$5.6 \times 10 \Rightarrow 3.59 \text{ k}\Omega$$

$$I_1 = I_T = \frac{10}{3.59}$$

$$I_T \Rightarrow 0.003 \text{ k}\Omega$$

$$I_2 = \frac{I_T \times R_1}{R_1 + R_2}$$

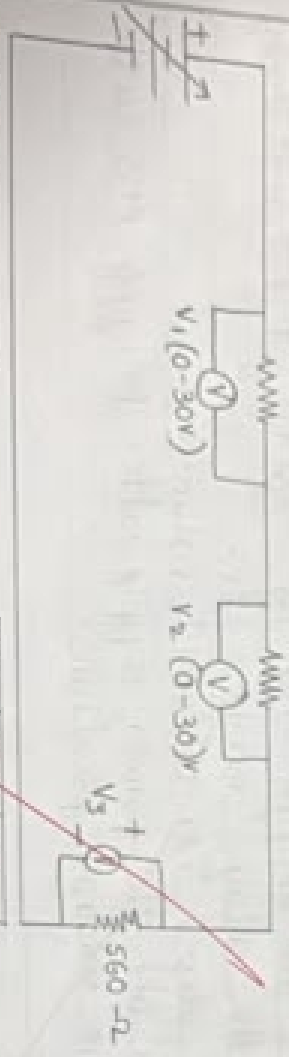
$$R_1 + R_2 = 10 + 5.73$$

$$I_2 \Rightarrow 0.00469$$

$$I_3 = \frac{I_T \times R_2}{R_1 + R_2} = \frac{0.003 \times 5.6}{10 + 5.73} = 0.0026$$

$$\Rightarrow I_2 + I_3 = I_1$$

$$0.00469 + 0.0026 \Rightarrow 0.0073$$



V	I ₁	I ₂	I ₃
10	0.34	0.48	0.26
15	1.11	0.77	0.39
20	1.49	0.95	0.53

V _{1h}	V ₂	V ₃
5	4.22	0.34
10	8.44	1.51
15	12.66	2.97

$$I_1 = I_2 + I_3$$

$$= 0.0048 + 0.0026 = 0.0074$$

$$= 0.0074 + 0.0039 = 0.011$$

$$0.011 \times 10 = 0.11 \text{ V}$$

$$V = V_1 + V_2 \Rightarrow 4.22 + 0.94 = 5.16$$

$$\Rightarrow 8.44 + 1.51 = 9.95$$

$$\Rightarrow 12.66 + 2.97 = 15.63$$

Exp. No. _____

Date _____ Page No. 044

Exp. No: 152 Verification of Current And Voltage Division Rules

Aim: To calculate the individual branch currents & total current drawn from the power supply using current and voltage division rules.

Apparatus Required:

S.No	Apparatus	Range	Quantity
1.	DC Regulated Power supply	(0-30)V	1
2.	Ammeter	(0-200)mA	4
3.	Resistor	1kΩ, 220Ω	each 2
4.	Bread board & wires	—	required

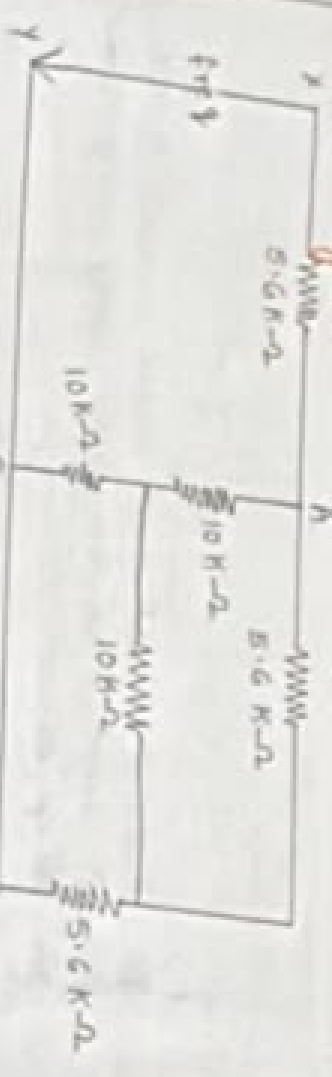
Procedure:

1. Give the connections as per circuit diagram.
2. Set a particular value in RPS.
3. Note down the corresponding ammeter reading.
4. Repeat the same for different voltages.

Result: Thus, the individual branch currents & total current drawn from power supply is successful.

Teacher's Signature _____

Circuit Diagram:



$$R_C = \frac{R_C R_E R_D}{R_C R_E + R_E R_D + R_D R_C} = \frac{12 \times 10^3 \times 490}{12 \times 10^3 + 490 + 10 \times 10^3} = 49.0 \Omega$$

$$R_D = \frac{R_C R_E R_D}{R_C R_E + R_E R_D + R_D R_C} = \frac{12 \times 10^3 \times 10 \times 10^3}{12 \times 10^3 + 490 + 10 \times 10^3} = 1646.9 \Omega$$

$$R_E = \frac{R_C R_E R_D}{R_C R_E + R_E R_D + R_D R_C} = \frac{12 \times 10^3 \times 490}{12 \times 10^3 + 490 + 10 \times 10^3} = 915.5 \Omega$$

$$R_D = 35.859 \Omega$$

$$R_D = 49.0 + 291.496 + 915.5 \Omega$$

$$R_D = 1646.9 \Omega$$

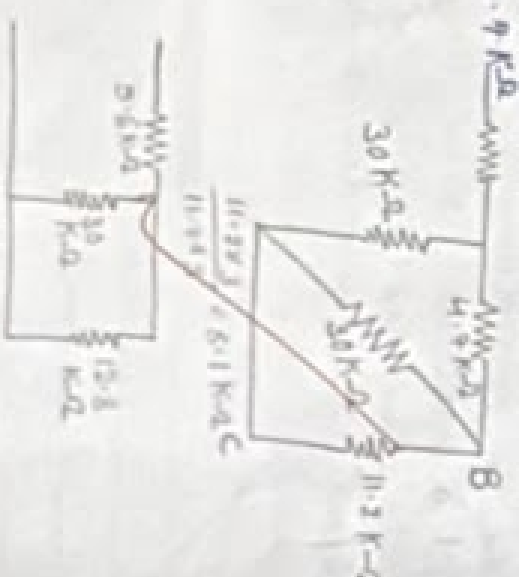
$$R_D = 1.647 K\Omega$$



$$R_{AB} = 30K + 5K + 4K = 39K\Omega$$

$$R_{BC} = 30K + 5K + 4K = 39K\Omega$$

$$R_{CA} = 30K + 5K + 4K = 39K\Omega$$



$$R_{AB} = 30K + 5K + 4K = 39K\Omega$$

$$R_{BC} = 30K + 5K + 4K = 39K\Omega$$

$$R_{CA} = 30K + 5K + 4K = 39K\Omega$$

Radical	Decomposed
1.67 KΩ	1.67 KΩ

Exp. No: 3

Verification of Star Delta Transformation Using Resistance Reduction Technique

Aim: To calculate the equivalent circuit resistance using star delta transformation technique.

Apparatus Required:

S.No.	Apparatus Name	Range	Quantity
1	Resistor		
2	Bread board & Connecting wires	-	Required

Procedure:

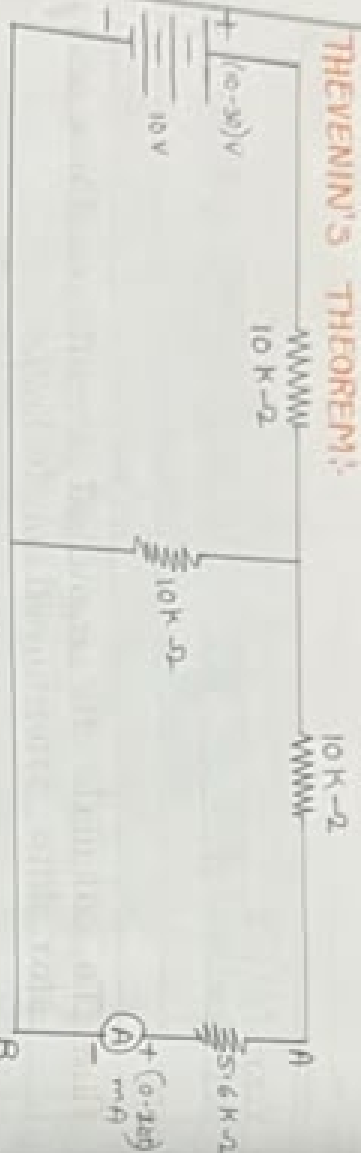
1. Give the connections as per the circuit diagram.
2. Determine the equivalent resistance of circuit between P and Q using star delta technique.
3. Verify the same by connecting multimeter across P-Q.

Result:

Thus, the equivalent circuit resistance is obtained using star delta transformation.

Teacher's Signature _____

THEVENIN'S THEOREM:



S.No	Vs	IL
1	10	0.24
2	20	0.49
3	30	0.73

$$V = IR$$

$$\Delta = \begin{vmatrix} 10 & -10 \\ 20 & -10 \end{vmatrix} = 10 \times 10 - 20 \times 10 = 100 - 200 = -100$$

$$I_1 = \frac{\Delta I_1}{\Delta} = \frac{100}{-100} = -1 \text{ A}$$

$$I_2 = \frac{\Delta I_2}{\Delta} = \frac{100}{-100} = -1 \text{ A}$$



S.No	Vs	Vth
1	10	4.9
2	20	9.9
3	30	14.9

$$V = IR$$

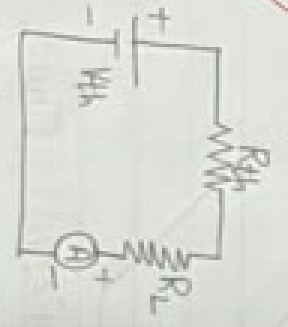
$$I = \frac{V}{R}$$

$$I = \frac{4.9}{10 \times 10^3} = 4.9 \times 10^{-4} \text{ A}$$

$$I = \frac{9.9}{20 \times 10^3} = 4.95 \times 10^{-4} \text{ A}$$

$$I = \frac{14.9}{30 \times 10^3} = 4.97 \times 10^{-4} \text{ A}$$

S.No	Vth	IL
1	4.9	0.24
2	9.9	0.48
3	14.9	0.73



S.No	Vs	Rth
1	10	14.64
2	20	14.64
3	30	14.64

$$I_L = \frac{V_{th}}{R_{th} + R_L}$$

$$I_L = \frac{4.9}{14.64 + 5.6} = 0.24 \text{ mA}$$

$$I_L = \frac{9.9}{14.64 + 5.6} = 0.48 \text{ mA}$$

$$I_L = \frac{14.9}{14.64 + 5.6} = 0.73 \text{ mA}$$

EXP. NO. 4 VERIFICATION OF THEVENIN'S AND NORTON'S THEOREMS

Aim: To verify the equivalent circuit parameters of Thevenin's and Norton's theorems theoretically and practically.

Apparatus Required:

S.No.	Apparatus Name	Range	Quantity
1.	Dc regulated power supply	(0-30) V	1
2.	Voltsmeter	(0-30) V	1
3.	Ammeter	(0-200) mA	1
4.	Resistor	330Ω, 220Ω, 100Ω, 560Ω, 100Ω	5
5.	Multimeter	—	1
6.	Bread board & connecting wires	—	—

Procedure for Thevenin's Theorem:

1. Give the connections as per the circuit diagram.
2. Measure R_{th} using a multimeter by killing sources and open circuit R_L .
3. Measure V_{th} across A & B.
4. Measure load current I_L through R_L .

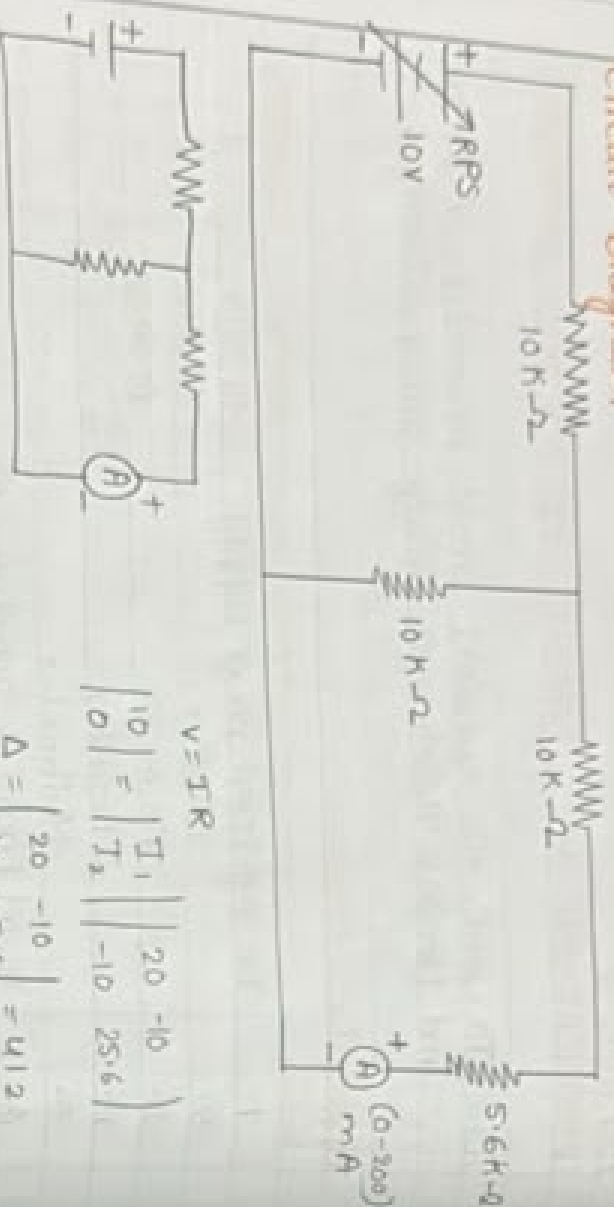
$$I_L = \frac{V_{th}}{R_{th} + R_L}$$

Procedure for Norton's Theorem:

1. Give the connections as per circuit diagram.
2. Measure R_{th} using a multimeter by killing sources.
3. Measure I_N through A and B.
4. Measure load current I_L through R_L .

Teacher's Signature _____

NORTON'S THEOREM: Circuit Diagram:



$$V = IR$$

$$\begin{vmatrix} 10 \\ 0 \end{vmatrix} = \begin{vmatrix} I_1 \\ I_2 \end{vmatrix} \begin{vmatrix} 20 & -10 \\ -10 & 25.6 \end{vmatrix}$$

$$\Delta = \begin{vmatrix} 20 & -10 \\ -10 & 25.6 \end{vmatrix} = 412$$

S.No.	V_s	I_L
1	10	0.24
2	20	0.49
3	30	0.73

S.No.	V	I_{sc}
1	10	0.33
2	20	0.67
3	30	1

$$I_2 = \frac{\Delta I_2}{\Delta}$$

$$\Delta I_2 = \begin{vmatrix} 20 & 0 \\ -10 & 0 \end{vmatrix} = 200$$

$$I_2 = \frac{100}{412} = 0.24$$

$$2. \Delta I_2 = \begin{vmatrix} 20 & 20 \\ -10 & 0 \end{vmatrix} = 0 + 200 = 200$$

$$I_2 = 0.49$$

$$3. \Delta I_2 = \begin{vmatrix} 20 & 30 \\ -10 & 0 \end{vmatrix}$$

$$= 0 + 300 = 300$$

$$I_2 = 300/412 = 0.73$$

S.No.	I_{sc}	I_L
1	0.33	0.24
2	0.67	0.49
3	0.73	1

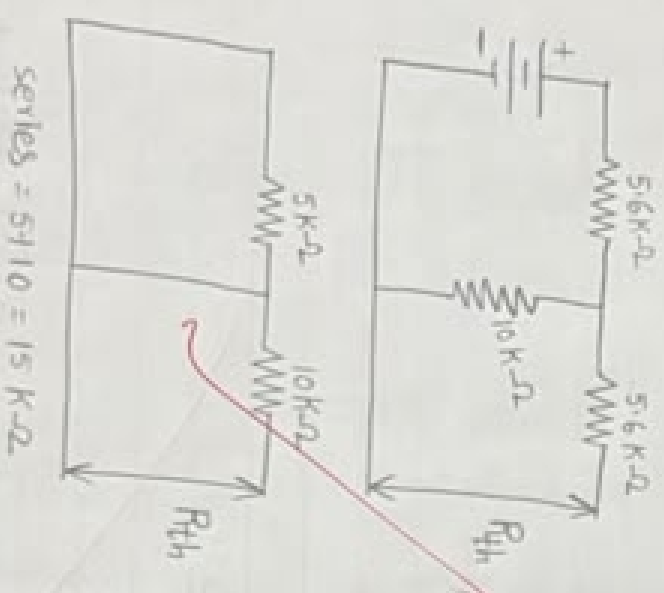
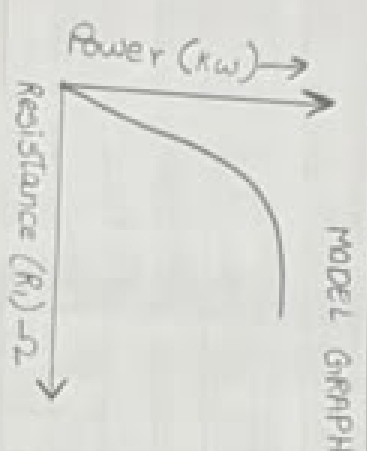
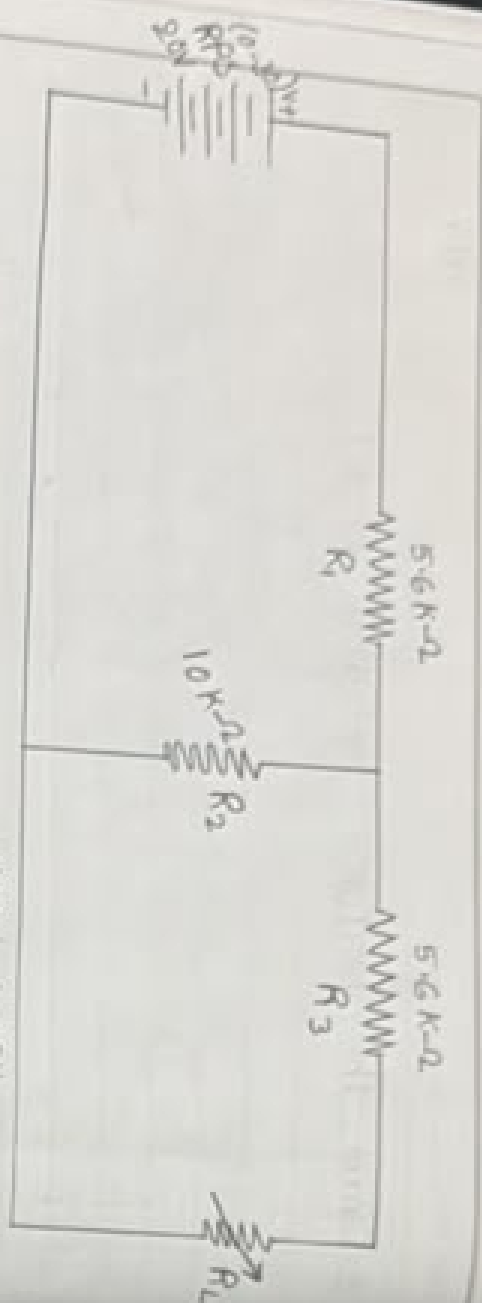


5. Draw the Norton's equivalent circuit.

Result: Thus, the equivalent circuit parameters are obtained using Thevenin's and Norton's theorem.

Teacher's Signature

S.No	R_L	I_L	P
1	1	0.47	0.22
2	2	0.45	0.4
3	3	0.42	0.52
4	4	0.40	0.64
5	5	0.38	0.72
6	6	0.37	0.77
7	7	0.35	0.80
8	8	0.33	0.87
9	9	0.32	0.86
10	10	0.30	0.90



Series = $5.6K + 10K = 15.6K - \Omega$

Exp. No. _____

Date _____
Page No. 08

EXP. NO. 5 VERIFICATION OF MAXIMUM POWER TRANSFER THEOREM AND SUPERPOSITION THEOREM

5.6) Aim: To verify maximum power transfer theorem.

Apparatus Required:

S.No.	Apparatus Name	Range	Quantity
1.	Dc regulated power supply	(0-30)V	1
2.	Voltmeter	(0-30)V	1
3.	Ammeter	(0-200)mA	1
4.	Resistor	300Ω, 2K-Ω	each 2
5.	Multimeter	—	1
6.	Bread board & Connecting wires	—	required

Procedure:

1. Give the connections as per circuit diagram.
2. Measure R_{th} using a multimeter.
3. Measure V_{th} across $200 - \Omega$ (R_2).
4. Measure load current I_L through R_L .
5. Calculate maximum power transferred to load.

Calculation:

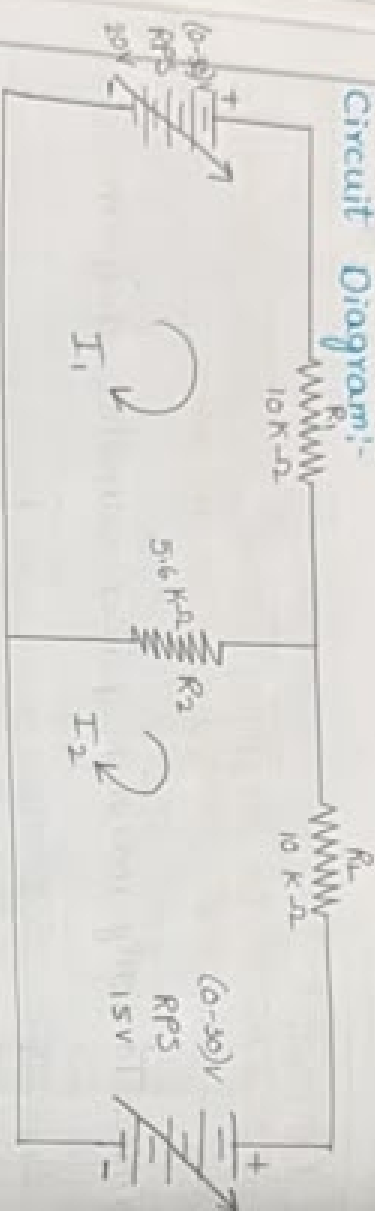
$$P = I_L^2 R_L$$

- (i) $(0.47)^2 \times 1 = 0.22$
- (ii) $(0.45)^2 \times 2 = 0.4$
- (iii) $(0.42)^2 \times 3 = 0.52$
- (iv) $(0.40)^2 \times 4 = 0.64$
- (v) $(0.38)^2 \times 5 = 0.72$

Result: Thus, the maximum power transfer theorem is verified.

Teacher's Signature _____

Circuit Diagram:-



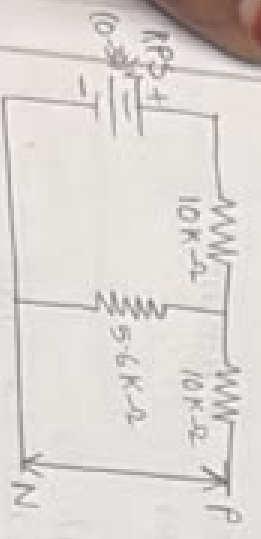
TRIBULAR FORM

S.No	V ₁	V ₂	I _L
1	10	10	0.94
2	15	20	2.14
3	20	30	2.37

S.No	V ₁	I _L
1	10	0.4
2	15	0.7
3	20	0.9

S.No	V ₁	I _L
1	10	0.47
2	30	1.42
3	30	1.42

Calculation:-



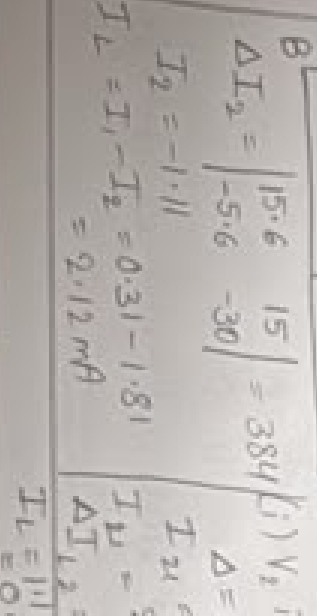
$$|R| \left| \begin{matrix} I_1 \\ I_2 \end{matrix} \right| = \begin{vmatrix} 15 \\ -30 \end{vmatrix}$$

$$A = \begin{vmatrix} 15.6 & -5.6 \\ -3.6 & 15.6 \end{vmatrix} = 2.12$$



$$\Delta I_1 = \begin{vmatrix} 15 & -5.6 \\ 30 & 15.6 \end{vmatrix} = 6.6$$

$$I_1 = \frac{\Delta I_1}{\Delta} = \frac{6.6}{2.12} = 0.31$$



$$\Delta I_2 = \begin{vmatrix} 15.6 & 15 \\ -5.6 & -30 \end{vmatrix} = 384$$

$$I_2 = \frac{\Delta I_2}{\Delta} = \frac{384}{2.12} = 180.66$$

5.b) Aim:- To determine the current flow through the load resistors using superposition theorem.

Apparatus Required:-

S.No.	Apparatus Name	Range	Quantity
1.	DC regulated power supply	(0-30)V	1
2.	Voltmeter	(0-30)V	1
3.	Ammeter	(0-200)mA	1
4.	Resistor	1kΩ, 220Ω, 56Ω	each 1
5.	Multi-meter	-	1
6.	Breadboard & Connecting wires	-	required

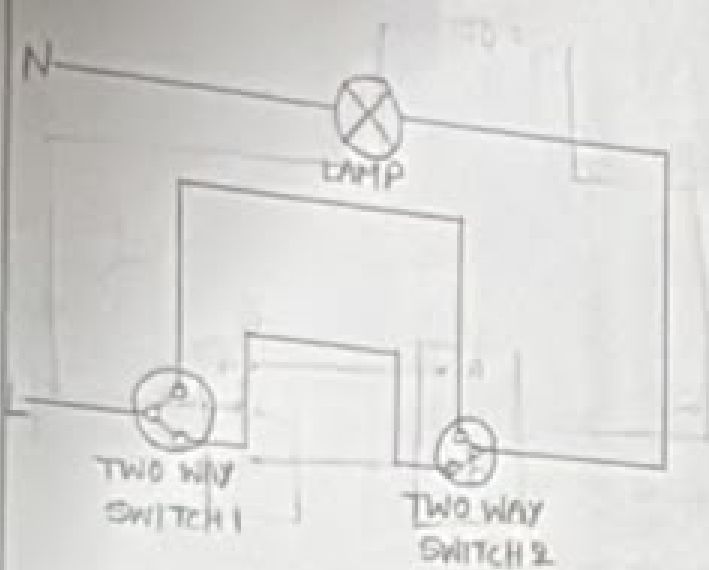
Procedure:-

1. Give the connections as per the circuit diagram.
2. Measure the current flow through 1kΩ by connecting both the supplies.
3. Short circuits 15V source.
4. Measure current flow through 1kΩ by connecting 20V supply.
5. Short circuit 20V source.
6. Measure current flow through 1kΩ by connecting 15V supply.
7. Verify the net current through 1kΩ resistor.

Result:- Thus, the current flow through load resistor is determined using superposition theorem.

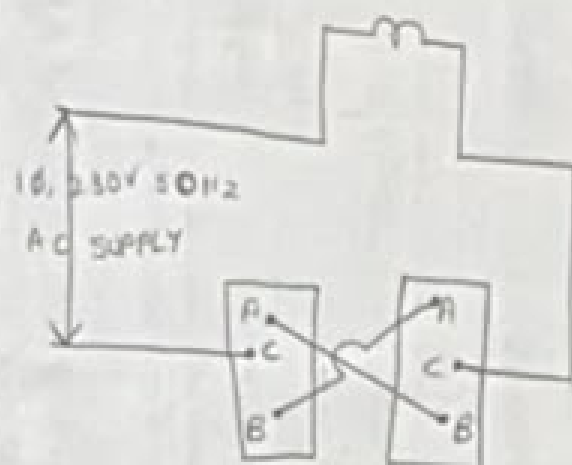
Teacher's Signature _____

Direct Connection



S. No.	S ₁	S ₂	Lamp status
1	CA	C'A'	ON
2	CB	C'B'	ON
3	CA	C'A'	OFF
4	CB	C'B'	OFF

INDIRECT CONNECTION



S. No.	S ₁	S ₂	Lamp status
1	CA	C'A'	OFF
2	CB	C'B'	OFF
3	CA	C'B'	ON
4	CB	C'A'	ON

Exp. No. 3

STAIRCASE WIRING AND INCANDESCENT TUBE WIRING

A. STAIRCASE WIRING

Aim: To control status of given lamp using 2 way switches

Apparatus Required:

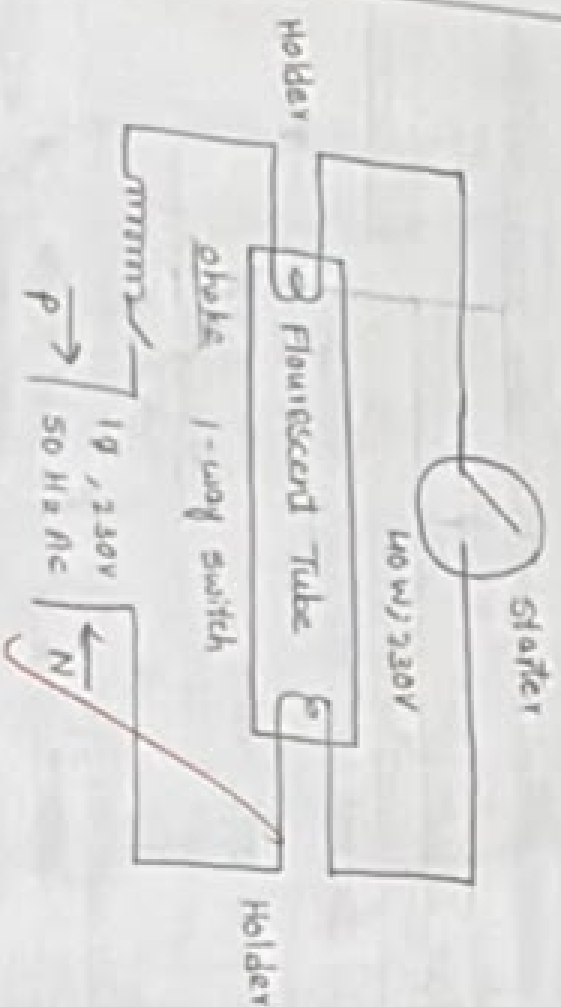
S. No.	Apparatus	Range	Quantity
1.	Incandescent lamp	60W/100W	1
2.	SPDT	5A, 250V	2
3.	Lamp holder	Rated type	1
4.	Line tester	500V/1000V AC	1
5.	3 pin plug	5A, 250V	1
6.	wire stripper/can cutter	AC 950	1
7.	Connecting wires	12A/15A mm ²	as required

Procedure:

1. A piece of wire is connected to the phase side and other end to middle point of SPDT.
2. Another point of lamp holder is connected to neutral line.
3. Upper point of SPDT switch 1 is connected to the upper point of SPDT switch 2. Lower point of SPDT switch 1 is connected to lower point of switch 2.
4. Circuit is tested that all combinations of switch connection.

Result:- Thus, the status of given lamp was controlled by and tested under direct and indirect connection.

Teacher's Signature



FLUORESCENT TUBE WIRING

Aim: To prepare wiring for a fluorescent tube light with switch control.

Apparatus Required:

S. No.	Apparatus	Range	Quantity
1.	Tube light with fitting	-	1
2.	Joint clips	-	as required
3.	Switch	-	1
4.	wires	-	as required
5.	screws	-	as required
6.	Switch board	-	1

Procedure:

1. Mark the switch and tube light location points and draw lines for wiring on wooden board.
2. Place wires along the lines and fix them with the help of clips.
3. Fix the switch and tube light fitting in the marked positions.
4. Complete the wiring as per the wiring diagram.
5. Test the working of the tube light by giving electric supply to the circuit.

Result: Thus, the wiring for the tube light is connected and tested.

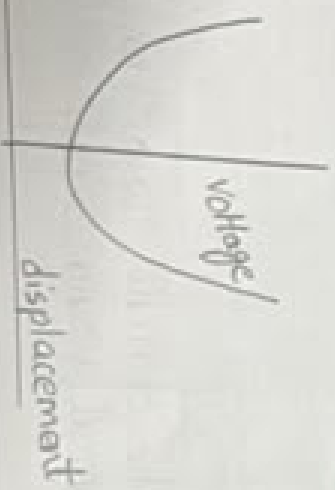
Teacher's Signature

Circuit Diagram:-



direction of displacement	displacement (mm)	Voltage (V)
-	0	5.99
-	3	4.13
-	6	2.30
-	8	0.95
-	8.5	0.52
-	9.5	0
-	10.5	-0.25
-	12	-1.78
-	13	-2.37
-	14	-3.08
-	15	-3.56

MODEL GRAPH:-



Exp. No. _____

Date _____
Page No. 10

Exp. No:-6

OUTPUT CHARACTERISTICS OF LVDT

Aim:- To plot the output characteristics of LVDT.

Apparatus Required:-

S.No.	Apparatus	Quantity
1.	LVDT kit	1
2.	Multimeter	1
3.	Connecting wires	As required

Procedure:-

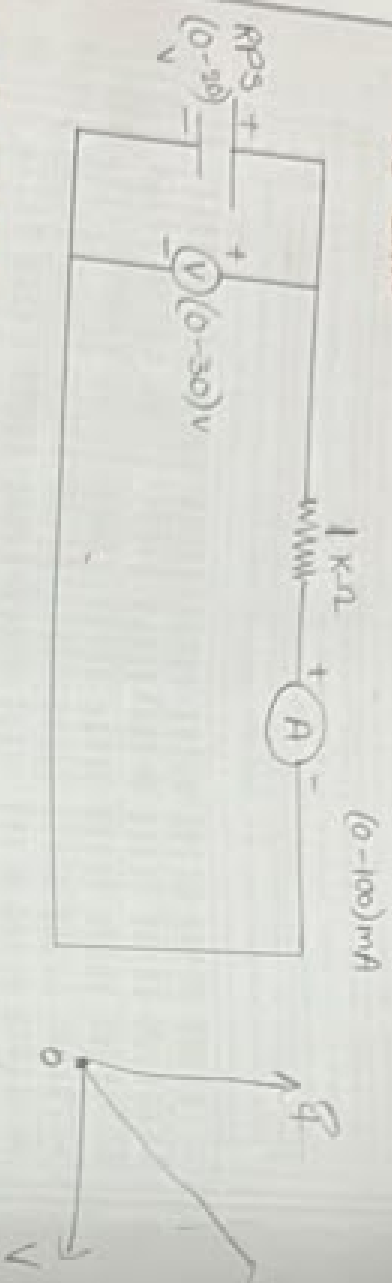
1. Connect the circuit according to circuit diagram.
2. Switch on the power supply.
3. The core is initially brought to null position.
4. First turn the core nut in clockwise direction to move the core towards right of null point and again take respective voltage readings from voltmeter.
5. Plot the graph from the observations taken.

Note

Result:- Thus, the output characteristics of LVDT is plotted.

Teacher's Signature _____

CIRCUIT DIAGRAM: OHM'S LAW:



TABULATION:

Supply Voltage (V)	Current (mA)	Constant	Theoretical constant
5.4	4	1kΩ	1.051k
9.9	8	1kΩ	1.125k
13.9	12	1kΩ	1.081k
18.2	16	1kΩ	1.041k
21.6	20	1kΩ	1.0k

Calculation:-

$$V = IR$$

$$R = \frac{V}{I} = \frac{V}{I \times 10^{-3}}$$

$$R_1 = \frac{5}{4} = 1.25 \text{ k}\Omega$$

$$R_2 = \frac{9}{8} = 1.125 \text{ k}\Omega$$

$$R_3 = \frac{13}{12} = 1.08 \text{ k}\Omega$$

$$R_4 = \frac{17}{16} = 1.06 \text{ k}\Omega$$

$$R_5 = \frac{20}{20} = 1 \text{ k}\Omega$$

Expt. No. _____

Date _____
Page No. 01

EXP. No: 1

VERIFICATION OF OHM'S LAW AND KIRCHHOFF'S LAW

1(a) Verification of Ohm's Law

Aim:- To verify Ohm's Law for a given resistive network.

Apparatus Required:

S.No.	Apparatus name	Range	Quantity
1.	RPS	(0-30)V	1
2.	Ammeter	(0-200)mA	1
3.	Voltmeter	(0-30)V	1
4.	Resistor	1kΩ	1
5.	Rheostat	300Ω/2A	1
6.	Bread board	—	required
	Connecting wires	—	

Procedure:-

1. Make the connections as per circuit diagram.
2. Switch on the power supply to RPS and apply a voltage (say 10V) and take the reading of voltmeter and ammeter.
3. Adjust the rheostat in steps and take down the readings of ammeter.
4. Plot a graph with V along x-axis and I along y-axis.
5. The graph will be a straight line which verifies ohm's law.
6. Determine the slope of the V-I graph - The reciprocal of the slope gives resistance of the wire.

Teacher's Signature _____